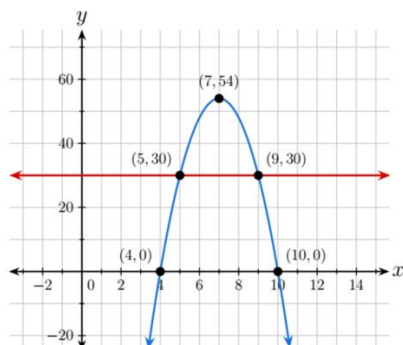


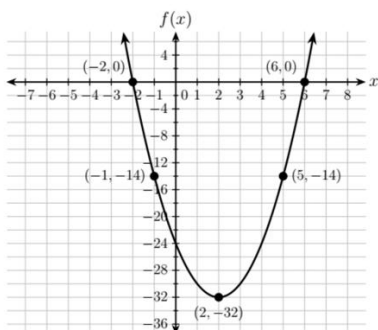
Algebra 1: Unit 9, Lesson 10

1. The equations $y = -6x^2 + 84x - 240$ and $y = 30$ are shown on the coordinate grid.



What are the solutions to the equation $-6x^2 + 84x - 240 = 30$?

2. Maria used a graphing utility to solve $2x^2 - 8x + 6 = 30$. Her graph of the quadratic function, $f(x) = 2x^2 - 8x - 24$, is shown.



What orders pairs shown on the graph are solutions for the quadratic $2x^2 - 8x + 6 = 30$?

3. Michelle is starting an ice cream selling business. She creates a function to help her determine what to charge customers per ice cream. The function is $P(x) = -3x^2 + 24x - 36$, where x is the cost of each ice cream in dollars and P is the profit in thousands of dollars.

Write an equation that Michelle should use to determine the cost of each ice cream when making a profit of \$8000.

